GSE 518
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Problem Set 5

You must turn in problems: 1, 2, 3 and 4(a). However, you are strongly encouraged to work on all the problems before the midterm, since the material they cover will be part of the midterm. Please let me know if you have any questions.

1. Let random variable X have a Normal distribution with mean $\mu = 1$ and variance $\sigma^2 = 4$. Use any standard normal tables to determine $\Pr(X < 1)$ and $\Pr(X \geq 1.5)$.

2. A small commuter airline uses planes with 20 seats. Its experience shows that 10% of individuals reserving space on a flight do not show up. If the company takes 23 reservations for each flight, what is the probability that it will be able to accommodate everyone appearing without bumping anyone? (*Hint*: Use a binomial distribution)

3. If the random variable X follows a Poisson distribution such that $\Pr(X = 1) = \Pr(X = 2)$, find $\Pr(X = 3)$.

4. (a) Show that if $X \sim Gamma(p/2, 2)$ and $Y \sim Chi\text{-}Squared(p)$, then $X \sim Y$. In other words, show that a $Gamma(p/2, 2)$ equals a $Chi\ Squared(p)$. (*Hint:* You do not need to derive the mgf's of these distributions, you can take those as given).

   (b) Show that if X follows a standard normal distribution $N(0, 1)$, then $Y = X^2$ follows a Chi-squared distribution with degrees of freedom equal to 1. (*Hint:* Follow steps similar to the ones in the example we did in class regarding Theorem 2.3.11.You can take the mgf of a Chi-squared distribution as given).

5. Let X be a discrete random variable with $\Pr(X = -1) = 1/8$, $\Pr(X = 0) = 6/8$, $\Pr(X = 1) = 1/8$ and $\Pr(X = a) = 0$ for all other values of a. Calculate the bound on $\Pr(|X - \mu| \geq k\sigma)$ for $k = 2$ using Chebyshev's inequality, where μ is the mean of X and σ its standard deviation. Compare this bound to the actual probability $\Pr(|X - \mu| \geq k\sigma)$. This is an example of a case in which Chebyshev's inequality gives a sharp bound.